

B.Tech. Examination-2024
Electronics and Communication Engineering
(Odd Semester Regular & Supplementary Examination)
Microprocessor and Microcontroller (ECEUGPC10)

F M: 80

Time: 3.00 Hrs

Answer Question No. 1 and any Seven (7) Questions from 2 to 12

Q. No.	Questions	M
1.	(a). Why the lower byte address bus (A0-A7) and data bus (D0-D7) are multiplexed?	2
	(b). Write a program to enable all interrupt except RST 5.5	2
	(c). Draw the flag structure of 8085 microprocessor.	2
	(d). Why program counter and stack pointer are 16 bits register of 8085 microprocessor?	2
	(e). Write a single instruction to clear the lower four bits of the accumulator in 8085 microprocessor.	2
	(f). Write a control word for mode 0 to configure port A and port CU as output port and port B & port CL as input port.	2
	(g). What is subroutine?	2
2.	(a). Draw and explain the software model of 8085 microprocessor.	5
	(b). Explain the function of demultiplexing circuit of 8085 microprocessors.	5
3.	(a). Explain the function of control and status signals of 8085 microprocessor.	5
	(b). Write down the function of the instructions a) LHL D b) DAA c) STAX d) INR M e) XRA M	5
4.	(a). Draw the timing diagram of MVI A, 08 (Opcode: 3E, 08) instruction stored from memory location 8000H.	5
	(b). Interface the 2KB RAM, and 2KB EPROM memory devices in 8085 microprocessor in absolute decoding method.	5
5.	(a). What are the interrupt pins of 8085 microprocessor? Explain briefly vector and non-vector interrupts of 8085. What is meant by priority interrupts?	5
	(b). How does an 8085 microprocessor respond to INTR interrupt signal?	5
6.	Eight BCD numbers are stored in consecutive memory locations starting from 8500 _H . Write a program to add them and store the result in 8600 _H and 8601 _H .	10
7.	(a). Write down the differences between memory mapped I/O and Peripheral mapped I/O.	5
	(b). Design a seven-segment common cathode LED output port with the device address FF _H and 8bit input port containing 8 number of switches with the device address FA _H . Write program to access the switches and display its value to LED.	5
8.	(a). Draw and discuss the functional block diagram of 8255.	5
	(b). Draw and explain the control word format of 8255.	5
9.	(a). Draw and discuss in brief the BIU and EU of 8086 microprocessor.	5
	(b). How is pipeline achieved in 8086 microprocessor?	5
10.	(a). Explain the concept of segmented memory? What are its advantages?	5
	(b). Explain the addressing modes of 8086 Microprocessor with proper examples.	5
11.	(a). Draw the schematic pin diagram of the 8051 microcontroller and explain operation of the following pins: (a) RxD, (b) XTAL2, (c) EA (d) PSEN	5
	(b). Draw and explain the Program Status Word of 8051 microcontroller.	5
12.	(a). Write down the differences between 8086 and 80286 microprocessor.	5
	(b). Draw the architecture of ARM Processor and explain why it is a RISC Processor.	5

TOTAL MARKS: 80

TIME: 3 Hours

INSTRUCTIONS: -

1. Mention the question number clearly. Answer all parts of a question at single location.
2. Draw block diagram & waveforms wherever necessary.
3. Acronyms & symbols have their usual meaning unless otherwise specified.
4. Make suitable assumptions wherever necessary.
5. Use separate answer scripts for Group A & Group B (ECE students only).

Group A

5×10=50

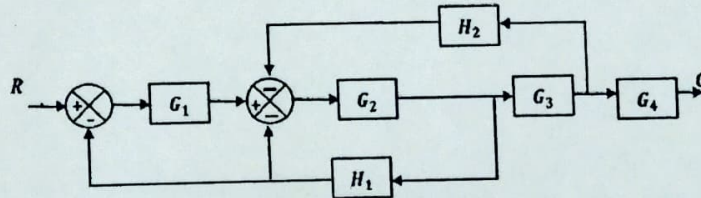
Answer Question-1 and any four from the rest.

Q-1 Answer any five questions.

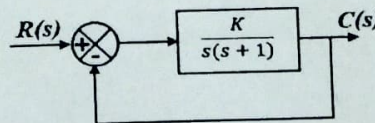
- (a) Explain the term feedback as applied to control systems. [2]
- (b) What is BIBO stability criterion? [2]
- (c) Give examples of open-loop and closed-loop control system. [2]
- (d) The transfer function of a system is $G(s) = \frac{100(s+5)(s+50)}{s^4(s+1)(s^2+2s+2)}$. Write type and order of the system. [2]
- (e) The expression of unit step response of a system is given by $c(t) = (1 - e^{-8t})$. Determine the transfer function of the system. [2]
- (f) What do mean by analogous system? In force-voltage analogy, mass is analogous to _____. [2]
- (g) What is quadrantal symmetry of a polynomial? [2]

Q-2 (a) Define linear system. [2]

- (b) Determine the overall transfer function (C/R) of the system shown in Fig. below. [8]

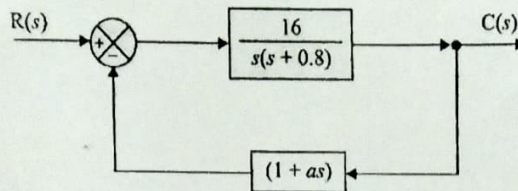


- Q-3 (a) The open-loop transfer function of a negative unity feedback system is $G(s) = \frac{144}{s(s+12)}$. Determine its transient response specifications for a unit step input. [6]
- (b) Determine the range of values K for the system shown in Fig. below so that steady-state error $e_{ss} < 0.004$ when $r(t) = 0.2t$. [4]



- Q-4 (a) Write the limitations of Routh's stability criterion. [2]
- (b) A unity feedback system has $G(s) = \frac{K}{s(s+1)(s+2)(s+4)}$. Find the following using Routh-Hurwitz's criterion: [8]
- i. the range of K for stability.
 - ii. the value of K for marginal stability and
 - iii. the poles on the imaginary axis of the s-plane (for marginal stability).

- Q-5 (a) Explain a DC potentiometer with a suitable diagram. [4]
- (b) Consider the system shown in Fig. below. Determine the value of a such that the damping ratio is 0.5. Also, obtain the values of the rise time and maximum overshoot in its step response. [6]



Q-6 (a) Draw the op-amp realization circuit of proportional-derivative (PD) controller and derive its transfer function. [4]

(b) Derive the transfer function $G(s) = \frac{\omega(s)}{E_a(s)}$ of an armature-controlled D.C. motor where $\omega(s)$ is the angular velocity of the rotor shaft and $E_a(s)$ is the external voltage applied to the armature winding of the motor. [6]

Q-7 A negative unity feedback control system has an open-loop transfer function $G(s) = \frac{K}{s(s^2 + 2s + 2)}$. [10]
Sketch the root locus of the system as K is varied from 0 to ∞ and determine the maximum value of K for stability.

Group B

3×10=30

Answer any three questions.

Q-8 Sketch the asymptotic Bode plot for the open-loop transfer function $G(s)H(s) = \frac{40}{s^2(s+4)(s+10)}$. Also [10]
determine its gain margin (GM), phase margin and stability of the closed-loop system (use semilog graph paper).

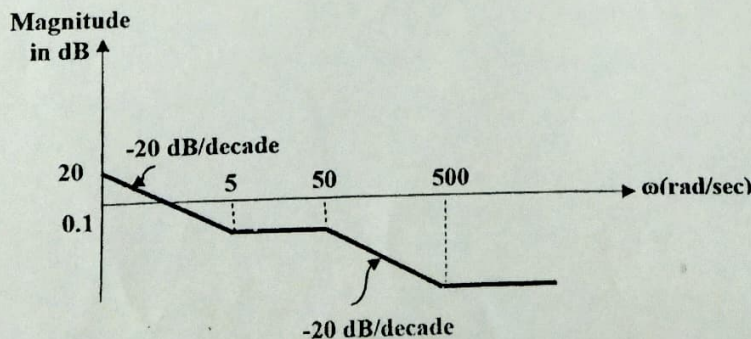
Q-9 Sketch the Nyquist plot and determine stability of the closed-loop system whose open-loop transfer [10]
function is $G(s)H(s) = \frac{12}{(s+1)(s+2)}$.

Q-10 (a) Define gain crossover frequency. [2]
(b) Consider the open-loop transfer function $G(s)H(s) = \frac{10}{s(s+4)(s+8)}$. Determine gain margin (GM) [8]
of the system.

Q-11 (a) A system is described by the following differential equation [5]
 $\frac{d^3y(t)}{dt^3} + 11\frac{dy(t)}{dt} + 5y(t) = 10u(t)$
where $y(t)$ and $u(t)$ are the output and input of the system respectively.
Represent the system by state-space model. [5]

(b) A system is described by the state equation
 $\dot{X}(t) = \begin{bmatrix} 0 & 1 \\ -1 & -2 \end{bmatrix} X(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t); y(t) = [1 \ 0] X(t)$
Determine the transfer function of the system. [2]

Q-12 (a) Write the advantages of Bode plot. [2]
(b) The asymptotic Bode magnitude plot of a system is given below. Determine its transfer function. [8]



- Instruction s:**
1. Use separate answer scripts for two groups.
 2. Mention the question number clearly and write all the parts of a question at one place.
 3. Write answers to the point, keeping in mind the allotted marks.
 4. Draw circuit/figure & waveforms wherever applicable (including numerical).
 5. Make suitable assumptions wherever necessary, symbols and notations have their usual meanings.

Group - A

Answer any Five (5 X 10 = 50 marks)

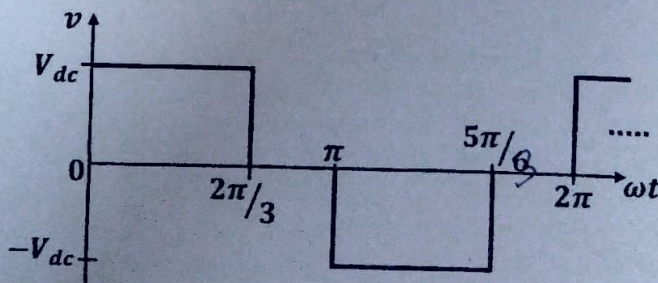
- Que-1** (A) Discuss the following terms for diodes: 6
✓ Softness factor, PIV, reverse recovery time.
- (B) Explain the effect of forward bias and reverse bias on the depletion layer in p-n junction diode. 4
- Que-2** (A) The beta (β) of bipolar transistor with common emitter connection varies from 10 to 60. The load resistance $R_L = 5 \Omega$. The dc supply voltage is $V_{CC} = 100V$ and the input voltage to the base circuit is $V_B = 8V$. If $V_{CE(sat)} = 2.5V$ and $V_{BE(sat)} = 1.75V$, find (a) the value of R_B that will result in saturation with an overdrive factor of 20; (b) the forced β , and (c) the power loss in the transistor P_T . 6
- (B) Why is it preferable to use hard drive for BJT? 2
- (C) What is difference between β over forced β_f for BJTs? 2
- Que-3** (A) Discuss how conduction takes place in n-channel enhancement type PMOSFET. Draw its transfer and output characteristics. 5
✓
- (B) A single-phase half-wave uncontrolled rectifier is connected to RL load. The supply voltage $V_s = 240V$ at 50Hz, $R = 10 \Omega$, $L = 5mH$, extinction angle is 210° . Find the average value of output voltage and output current. 5
- Que-4** Explain with the help of circuit diagram and output voltage characteristics the working of a single-phase full wave diode bridge rectifier an R- load. Draw the input/output voltages, output currents and diode voltages (use graph paper). Derive the expression of average and r.m.s. value of output voltage. 10
✓
- Que-5** (A) Explain the working of a single-phase full-wave controlled mid-point type rectifier circuit with R-L load. Draw the output voltage, diode voltage, output current waveforms. Derive the expression of average output voltage. 7
✓

- (B) Compare the mid-point type and bridge-type rectifier circuit. 3
- Que-6 (A) What is meant by step-down chopper? Explain its operation. Sketch the input voltage, input current, output voltage and output current waveforms. State the various assumptions made. 6
- (B) A step-up chopper with a pulse width of $100\mu s$ is operating from a 240V, 50Hz supply. Compute the output voltage for a chopping frequency of 2200Hz. 4
- Que-7 (A) Describe the working of a half-wave phase-controlled ac voltage controller with proper circuit diagram and output waveform. Derive the rms value of the output voltage. What are the limitations of half wave ac voltage controller. 5
- (B) Briefly explain the four possible configurations of single-phase ac voltage controllers. 5

Group - B

Answer any three ($3 \times 10 = 30$ marks)

- Que-7 Explain the working of a Single-Phase Half Bridge Inverter with RL-load. Derive the expression for the output voltage. 10
- Que-8 (A) A single-phase multiple pulse-width modulation bridge inverter controls the power in a resistive load. This inverter is driven by a 220 V dc supply. The trigger circuit for this control consists of a comparator that is fed from a 600 Hz carrier frequency and 50 Hz reference frequency. The width of each pulse is 18° . 5
- (a) Calculate the number of pulses per half cycle and the RMS value of the output voltage.
- (b) For the output voltage found in part (a), what will be maximum possible input voltage if the maximum pulse-width is given as 24° ?
- (B) Draw and explain the symbol and V-I characteristics of DIAC, TRIAC and GTO. What type of applications they can be used in? 5
- Que-9 For the given periodic waveform, find the THD using Fourier series expression. 10



- Que-10 Draw and explain the operation table (table indicating the conducting switches, phase voltages & line voltages in each of the six modes) for a three-phase inverter in 120° mode of operation. Using graph paper, draw the gate currents and phase and line voltages for the said inverter. 10

B. Tech. Examination-2024-25
Department of Electronics and Communication Engineering
(Odd Semester Regular & Supplementary)

Course Title: Micro and Nano Fabrication Process

Course Code: ECEUGPC12

Full Marks: 80

Time: 3.00 Hrs

- Answer all parts of a question in same place.
- Figures on the right hand side margin indicate full marks.
- Symbols & Acronyms have their usual meaning

Answer question no. 1, 2 and any 6 questions from group C

Group A (Answer any five questions 5x2=10)

- | | | | |
|----|-----|--|---|
| 1. | (a) | Mention two applications of GaAs | 2 |
| | (b) | Write application of single crystal Si | 2 |
| | (c) | What is wafer? | 2 |
| | (d) | Define twin boundary. | 2 |
| | (e) | What is frenkel defect? | 2 |
| | (f) | Define unit-cell. | 2 |
| | (g) | What is twin-tub process? | 2 |
| | (h) | Define Noise Margin? | 2 |

Group B (write two short notes 2x5=10)

- | | | | |
|----|-----|--|---|
| 2. | (a) | Poly-crystal Silicon | 5 |
| | (b) | Oxidation process | 5 |
| | (c) | CMOS inverter threshold voltage tuning to $V_{dd}/2$ | 5 |
| | (d) | Hot carrier effect | 5 |

Group C (Answer any six questions 6x10=60)

- | | | | |
|----|-----|--|---|
| 3. | (a) | Classify substrate material with examples | 4 |
| | (b) | Describe elemental, compound and Alloy Semiconductor with examples | 6 |
| 4. | (a) | Draw a tetrahedral structure of Si | 2 |
| | (b) | Draw a Diamond Lattice structure of Si crystal | 2 |
| 5. | (a) | with suitable diagram, Describe Czochralski (CZ) process to produce Si-ingot | 8 |
| | (b) | What is wafer? | 2 |

- | | | | |
|-----|-----|---|---|
| 6. | (a) | Classify 2D defects in crystal with suitable figure. | 4 |
| | (b) | A crystal plane intercepts x, y, z axis at 1, 2, 3 units. Calculate the crystal plane. | 3 |
| | (c) | Draw the common crystal planes of Si substrate | 3 |
| 7. | (a) | Describe ion-implantation process with a setup diagram. | 5 |
| | (b) | Compare between diffusion and ion-implantation process | 5 |
| 8. | (a) | What is photo resist? | 2 |
| | (b) | Classify photo resist | 2 |
| | (c) | Using different photo resist, explain SiO ₂ patterning process with suitable diagrams. | 6 |
| 9. | (a) | What materials are deposited using thin film deposition techniques? | 2 |
| | (b) | What is epitaxy? | 3 |
| | (c) | Describe Evaporation process with suitable diagram. | 5 |
| 10. | (a) | Using suitable figure, explain fabrication process of SOI wafer. | 5 |
| | (b) | Discuss Advantages of SOI technology | 5 |
| 11. | (a) | With mathematical deductions prove the four advantages of device Scaling | 8 |
| | (b) | Classify Device Scaling. | 2 |
| 12. | (a) | Draw NMOS structural diagram and explain the operation of MOS structure. | 6 |
| | (b) | Formulate the threshold voltage of NMOS | 4 |

-End-

B.Tech. 3 rd Year 5 th Semester Electronics and Communication Engineering

Sub: Object-oriented programming

Subject Code: CSEUGOE03

Time: 3 hrs; F.M. – 80.

Group- A (Marks distribution is shown in square bracket)

1. Answer the following questions (Each question contains 1 mark) [10x1 = 10]

- (i) What are the two ways in which Thread can be created in Java?
 - (ii) What do you mean by data type casting?
 - (iii) Define an abstract method 'sports' using Java.
 - (iv) Write two differences between method overloading and method overriding in Java.
 - (v) What do you mean by 'abstract' class in Java?
 - (vi) What is 'encapsulation' in OOP?
 - (vii) What is 'recursion' in programming?
 - (viii) How do we handle 'exceptions' in Java?
 - (ix) Write a Java programme to check whether 31 is even or odd.
 - (x) Why java is a platform independent programming language?
-

Group – B (Answer any five questions) [5x6 =30]

2.

- (i) How to implement thread by extending the 'Thread' class, show with a Java programme.
- (ii) Use 'while' keyword to print ten whole numbers between 10 and 21 using Java. [4 + 2]

3.

- (i) What do you mean by 'inheritance' in OOP? Explain inheritance with a proper example using Java.
- (ii) What do you mean by 'polymorphism' in OOP? [5 + 1]

4.

- (i) When does Java throw an 'ArrayIndexOutOfBoundsException'? How can we handle this exception? Show with an example.
- (ii) Write a Java programme to store 10, 20, 30, 40 and 50 in an array. [5+1]

5.

- (i) Write a Java programme to find the factorial of a given number.
- (ii) Write a Java programme to print first five positive odd numbers using 'for' loop. [4+2]

6.

- (i) What are the different types of access modifiers in Java? What is the difference between 'public' and 'default' access modifiers?
- (ii) Write a Java program to check whether a given number is a factor of 3. [2+ 4]

7.

- (i) A Java array contains the following elements: 'Kolkata', 'Delhi', 'Chennai', 'Bangalore', and 'Mumbai'. Use a for loop to display these elements on the console.
- (ii) Write a Java recursion to find the difference between the sum of the first twenty whole numbers and the first ten whole numbers. [2 + 4]
-

Group-C (Answer any four questions) [4x10 =40]

8.

- (i) Write a Java programme to check whether a number is palindrome or not.
- (ii) Write a Java recursion to find the sum of the series: $1 + 2 + 3 + \dots + 10$. [5 + 5]

9.

- (i) Write a Java programme to check whether a number is prime or non-prime.
- (ii) Write a Java programme to create the 'Employee' and 'Student' classes in package 1. The Employee class has two attributes i.e., 'emp_id' and 'salary'. Value for 'emp_id' and 'salary' respectively are '200501' and '50000'. The 'Student' class has a void type function called 'verify', which displays 'Authentic student of this university'. Now, create a new class called 'University' in package 2 and access the 'Employee' and 'Student' classes to display all of the items from those classes. [4+6]

10.

- (i) Write a Java programme to find the maximum and the minimum element of an array.
- (ii) Show method overriding in polymorphism with proper example using Java. [5+5]

11.

- (i) How 'abstract' class and methods are implemented in Java, show with an example.
- (ii) What is constructor in Java? Explain with a proper example. [5+5]

12.

- (i) Write a Java programme to find the sum of all of the digits of a number.
- (ii) Use binary search to find an item from a sorted array using Java. [5+5]

B.Tech. Examination-2024-25
Department of Electronics and Communication Engineering
(Odd Semester Regular & Supplementary)
Course Title: Communication System, Course Code: ECEUGPC11

Time: 3.00 Hrs

Full Marks: 80

- Answer all parts of a question in same place.
 - Figures on the right hand side margin indicate full marks.
 - Symbols have their usual meaning
- Answer any 5 from Question No. 1 and any 7 from Question No. 2 to 12 Questions**

- | | | |
|----|--|---|
| 1. | (a) What is baseband transmission? | 2 |
| | (b) What is modulation? | 2 |
| | (c) Define transmission efficiency. | 2 |
| | (d) What is frequency deviation in FM? | 2 |
| | (e) Write the properties of line coding. | 2 |
| | (f) What is lossless channel? | 2 |
| | (g) State Shannon-Hartley Law. | |
| 2. | (a) Draw the block diagram of a communication system and briefly explain the function of each block. | 5 |
| | (b) Mention different types of modulation. | 2 |
| | (c) Why do we need modulation in communication? | 3 |
| 3. | (a) What is meant by the term amplitude modulation? | 2 |
| | (b) Define modulation index for AM wave. | 4 |
| | (c) Derive an expression for single-tone amplitude modulated wave. | |
| | (d) Write the condition for over modulated AM wave and draw the waveform of over modulated AM wave. | 2 |
| 4. | (a) Prove that the balanced modulator produces an output signal consisting of two side band only with the carrier removed. | 5 |
| | (b) How is a SSB-SC signal demodulated? | 3 |
| | (c) Calculate the percentage of power savings for DSB-SC signal for the percentage modulation of 50% | 2 |
| 5. | (a) What is Carson's rule? | 2 |
| | (b) Explain the generation of narrowband FM signals with suitable block diagram. | 4 |
| | (c) A single tone frequency modulated wave is denoted by following expression $s(t) = 2 \cos[2\pi \times 10^6 t + 10 \sin(2\pi \times 10^3 t)]$. Determine carrier frequency, modulating frequency, modulation index and maximum frequency deviation. | 4 |
| 6. | (a) Explain the operation of balanced slope detector. | 5 |
| | (b) What is Pre-emphasis? | 2 |
| | (c) Compare between FM and AM. | 3 |
| 7. | (a) State sampling theorem. | 2 |
| | (b) What is aliasing? How is it prevented? | 2 |
| | (c) Explain flat top sampling technique | 4 |

- (d) Find the Nyquist rate and the Nyquist interval for the signal
 $x(t) = \cos(4000\pi t) \cos(1000\pi t)$ 2
8. (a) How is an analog signal converted to a digital signal in PCM system? 5
 (b) Derive the expression for transmission bandwidth in a PCM system. 3
 (c) A television signal having a bandwidth of 4.2 MHz is transmitted using binary PCM system. Give that number of quantization level is 512. Determine transmission bandwidth and output signal to quantization noise ratio. 2
9. (a) Explain the operation of a delta modulator with suitable block diagram. 6
 (b) A delta modulation system is designed to operate at 5 times the Nyquist rate for a signal with 3 KHz bandwidth. The quantizing step size is 250 mV. Determine the maximum amplitude of a 2 KHz input sinusoidal signal for which delta modulator does not show slope overload distortion. 4
10. (a) Consider the binary data sequence 1011101011. Draw the waveform for the following signalling format: (i) Bipolar RZ (ii) Split Phase Manchester coding (iii) AMI 3
 (b) Sketch the waveform of ASK, FSK and PSK signal for the input binary sequence 1101010010. 3
 (c) Explain briefly the coherent detection of BFSK signal with suitable block diagram. 4
11. (a) Define entropy, information rate and write their relation. 3
 (b) A DMS X has five symbols x_1, x_2, x_3, x_4 and x_5 with $P(x_1) = 0.4, P(x_2) = 0.19, P(x_3) = 0.16, P(x_4) = 0.15$ and $P(x_5) = 0.1$. Construct a Huffman Code for X. 4
 (c) A voice grade channel of the telephone network has a bandwidth of 3.4 kHz. Calculate the information capacity of the telephone channel for a signal-to-noise ratio of 30 dB. 3
12. (a) Given a (7,4) linear block code whose generator matrix is given by :

$$G = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

 (i) Find codeword corresponding to message word 0 1 0 1 (ii) Find the parity check matrix. Decode the received codeword 1 0 1 0 1 0 1. 6
 (b) The generator polynomial for a (7, 4) cyclic Hamming Code is given by, $G(p) = 1 + p + p^3$. Determine the systematic and non-systematic codeword for message word 1 0 1 0. 4

-End-